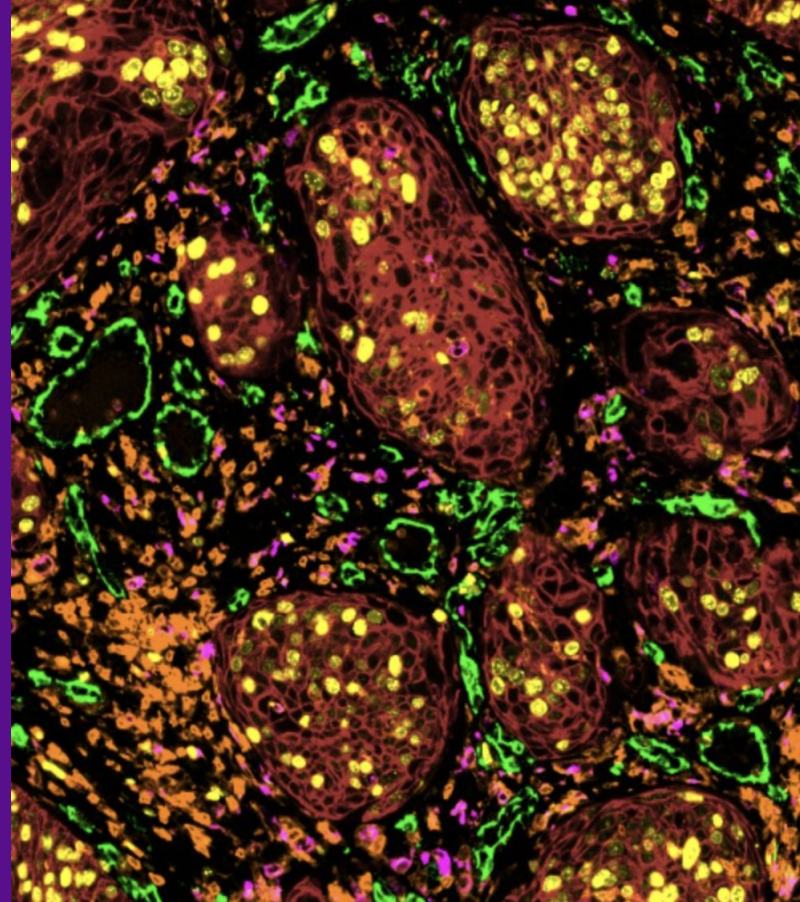


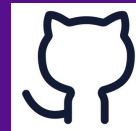


HUPO 2026 Bioinformatics Hub:

Strategies for Extracting Robust Biological Insights from Imaging-based Spatial Proteomics Studies



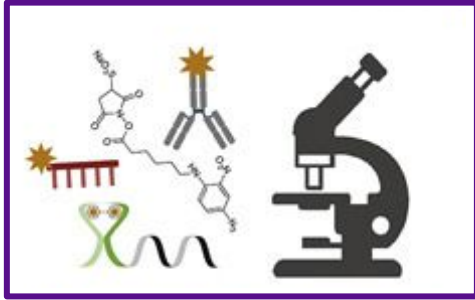
NYU Grossman School of Medicine



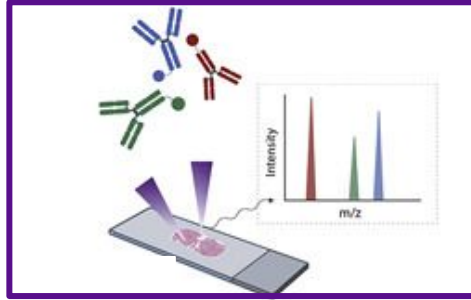
[github.com/FenyoLab/
hupo26-spatial-proteomics-hub](https://github.com/FenyoLab/hupo26-spatial-proteomics-hub)

Technologies of imaging-based spatial proteomics

Targeted

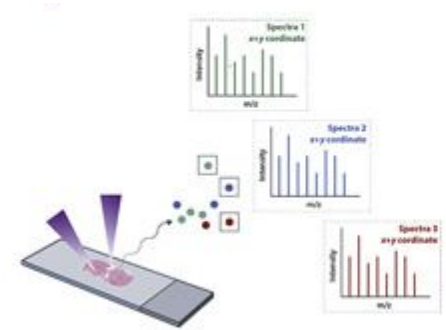


Labeling with affinity reagents/fluorescent proteins
(e.g., CODEX and COMET)



Imaging mass cytometry

Untargeted



Mass spectrometry imaging

Our lab's focus: characterizing the tissue microenvironment

Segmentation-based approach

Cells

Relies on accurate cell-segmentation algorithm

Can work with smaller datasets, though this may limit statistical power

Cell morphologies are discarded/summarized into a shape, and marker expressions are assigned to cells

More straightforward and easier implementability (cell-centric interpretation is intuitive)

Segmentation-free approach

Image pixels / block of pixels

Learns directly from raw data (often via self-supervised learning); the training has high computational costs

Large amounts of data is required

Retains cell morphology and assesses marker distribution without segmentation bias

More “black box”: learned signatures are harder to interpret and may be harder to translate into clinical routine

Bottleneck: lack of consistent analysis strategies

Distinct challenges in proteomics compared to transcriptomics →

Need for new analysis tools →

Need for ground-truth datasets & benchmarks

Analytical: dynamic range differences, multiplexing constraints, antibody specificity problems, ...

Biological: secreted proteins, PTMs, protein-protein interactions, ...

Synthetic data is a cost-effective way to generate customizable ground-truth datasets

Talk 3

Clinical applications of segmentation-based spatial proteomics

Characterizing the spatial immune landscape of mismatch repair deficient endometrial cancer

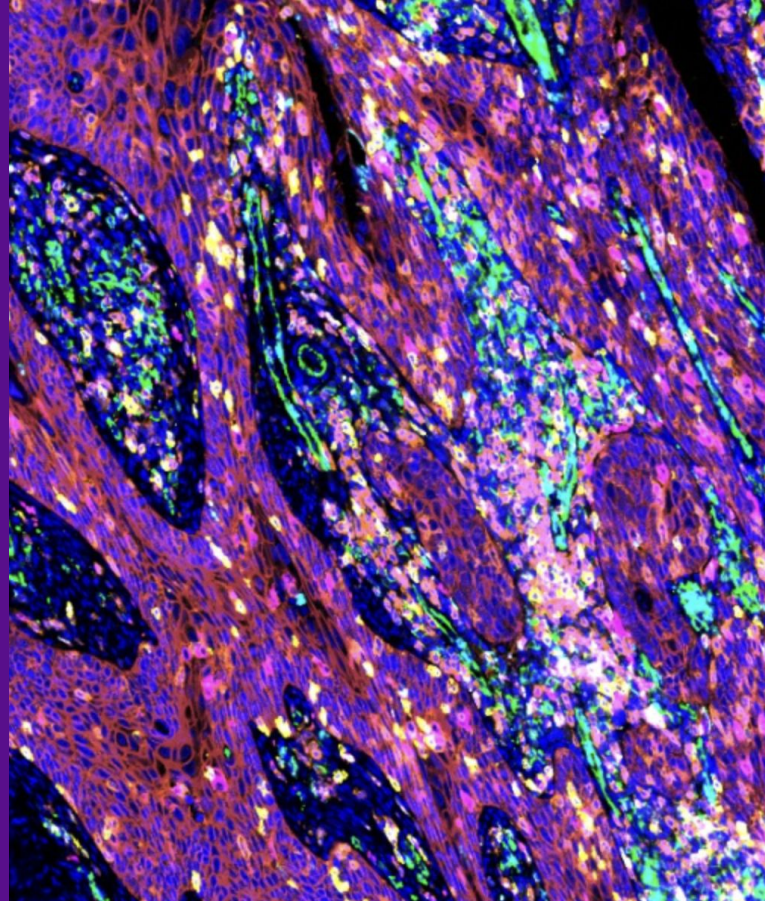
Michelle Hollenberg, PhD

Institute for Systems Genetics

Department of Biochemistry & Molecular Pharmacology



github



Section presented by Linda Procell

Our lab's focus: characterizing the tissue microenvironment

Segmentation-based approach

Cells

Unit of analysis

Relies on accurate cell-segmentation algorithm

Preprocessing

Can work with smaller datasets, though this may limit statistical power

Data requirements

Cell morphologies are discarded/summarized into a shape, and marker expressions are assigned to cells

Biological information retained

More straightforward and easier implementability (cell-centric interpretation is intuitive)

Interpretability / practical usability

Segmentation-free approach

Image pixels / block of pixels

Learns directly from raw data (often via self-supervised learning); the training has high computational costs

Large amounts of data is required

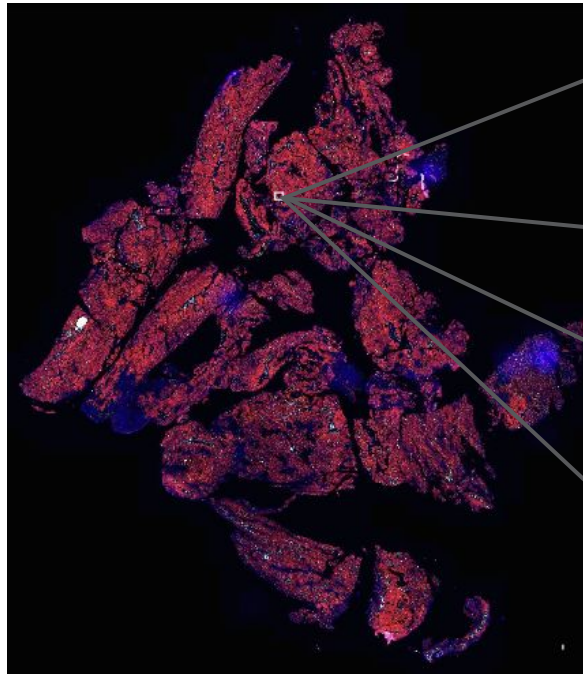
Retains cell morphology and assesses marker distribution without segmentation bias

More “black box”: learned signatures are harder to interpret and may be harder to translate into clinical routine

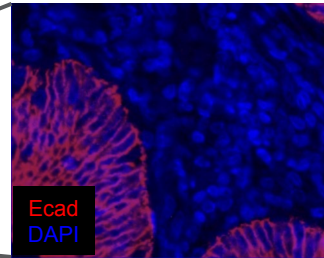
Segmentation converts pixels to biologically meaningful cells

Multiplexed IF Image

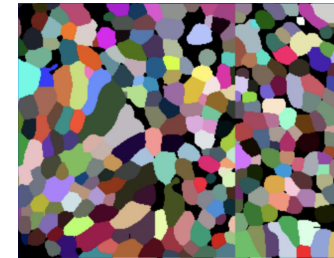
Whole Tissue - Pixel Level Data



1. Cell Segmentation

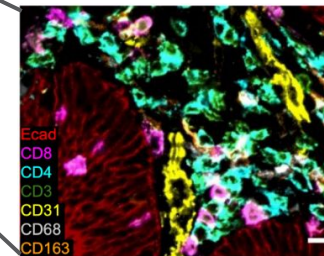


Nuclear & Tumor Cell
Membrane Markers

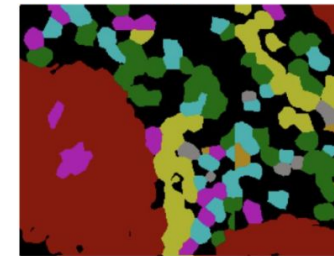


Cell Boundaries

2. Cell Type Identification



Cell Type Markers

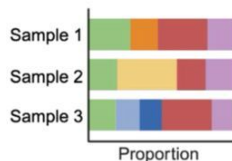


Cell Types

Segmented data analysis strategies lead to interpretable results

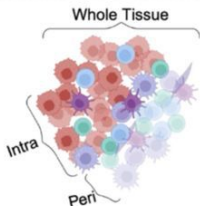
Calculate Per-Sample Cell Type Metrics by Region

Cell type proportions



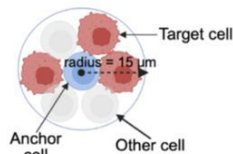
Fraction of intratumoral cells

$$\frac{\# \text{ intratumoral cells}}{\# \text{ cells in whole tissue (intra + peri)}}$$



Target cell enrichment

$$\frac{\# \text{ targets } \leq 15 \mu\text{m from anchor}}{\text{total } \# \text{ cells in } 15 \mu\text{m neighborhood}}$$

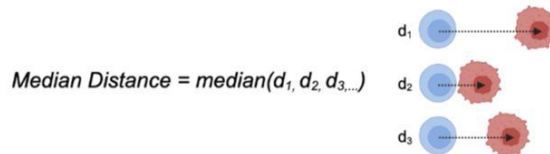


Anchor cells with neighboring targets

$$\frac{\# \text{ anchors } \leq 15 \mu\text{m from target}}{\text{total anchor cells}}$$



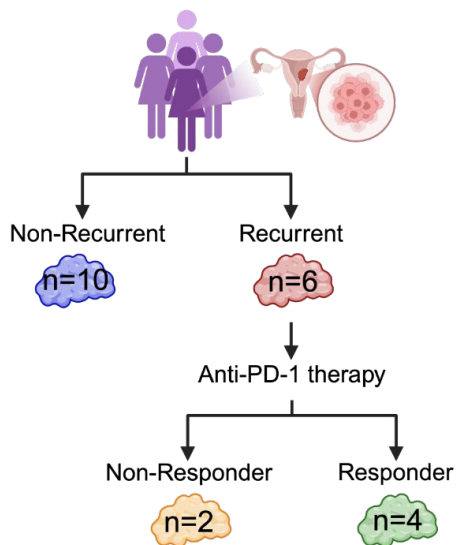
Median distance between anchor and closest target cell



Interpretable segmentation-based results benefit clinical studies: An example case

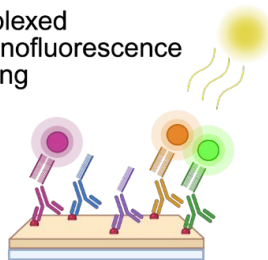
Study Design

16 Treatment-Naive
Primary dMMR
Endometrial Tumors

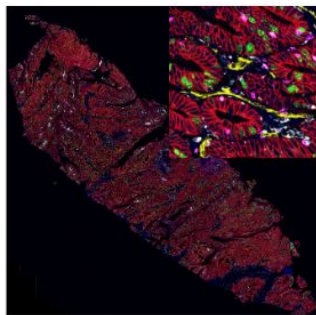


Data Generation

Multiplexed
Immunofluorescence
Imaging



Representative Image

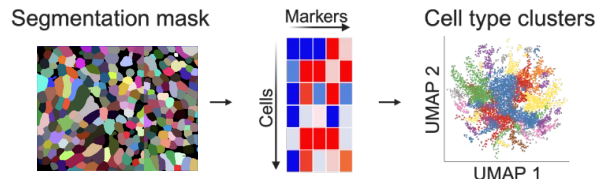


Computational Analysis and Biomarker Discovery

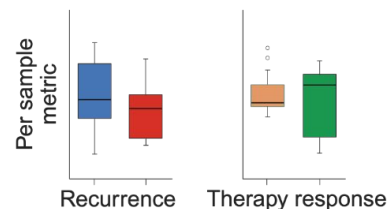
1. Image processing

- Tissue selection
- Tissue artifact removal
- Marker quality control

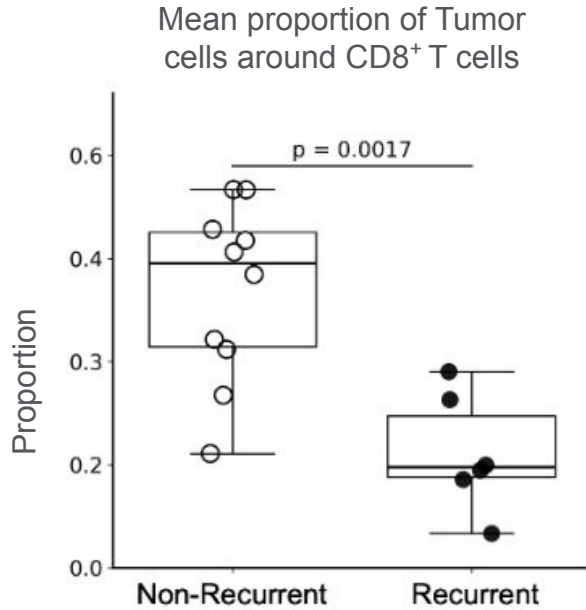
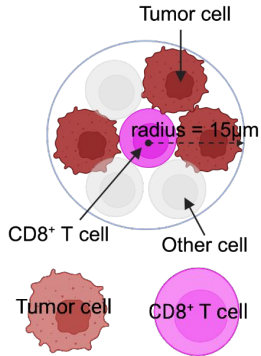
2. Cell segmentation and cell phenotyping



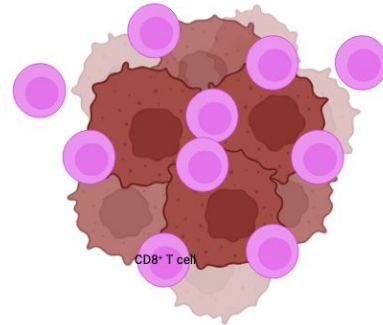
3. Correlation of cell type proportions and spatial metrics with clinical outcomes



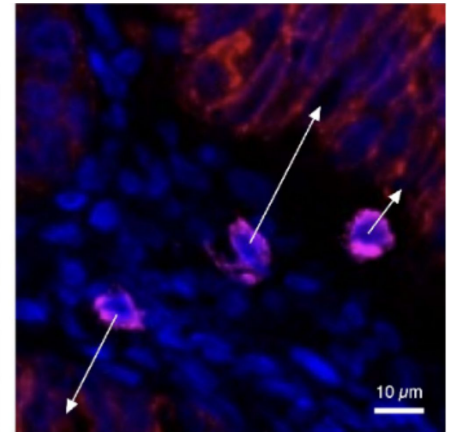
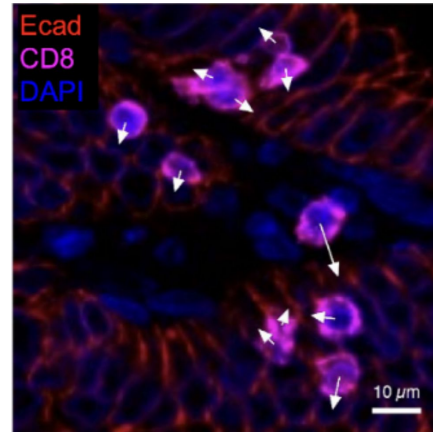
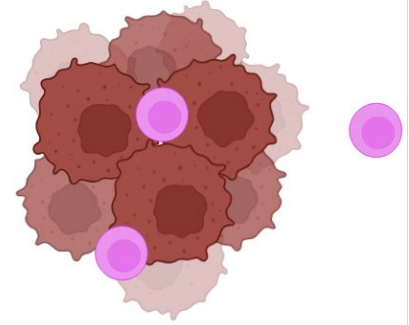
Tumor cells are more enriched around CD8⁺ T cells in non-recurrent tumors



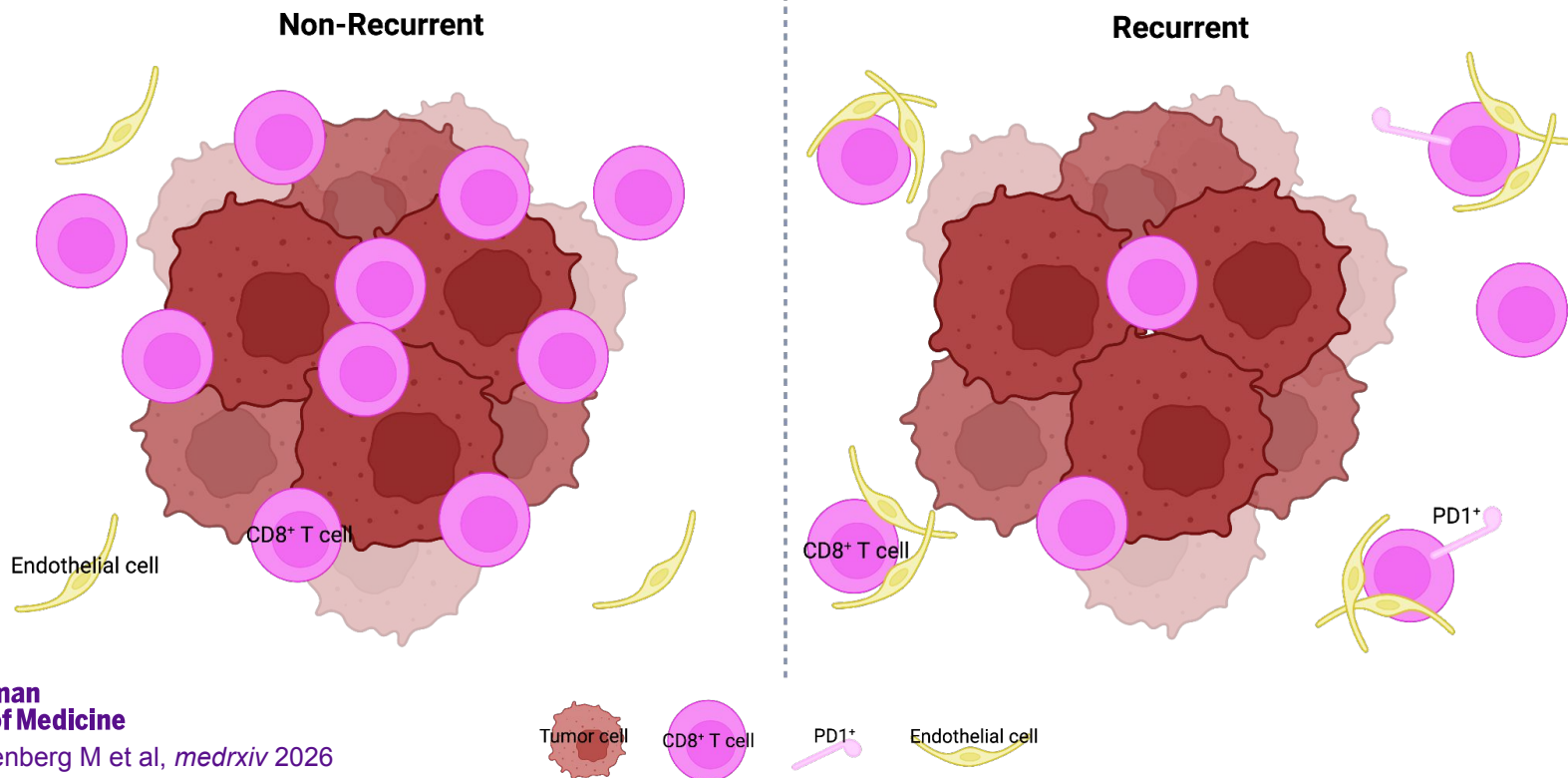
Non-Recurrent



Recurrent



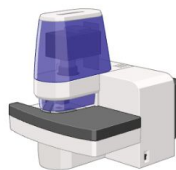
Non-recurrent tumors are more immune engaged; recurrent tumors are more immune excluded



Study-to-study data variation in spatial proteomics analysis

- Data limitations to consider:

Acquisition methods



Panel differences

Immune markers	
T cells	CD3e CD4 CD4 T FoxP3 TIM3 CD8 CD8 T Granzyme B Tox CD68 CD163 CD31 CD34 CD20 MPO CD45RO
Treg	Th1/Other
Cytotoxic	Exhausted
Macrophages	M2-like
Endothelial	
B cells	
Neutrophils	
	Activated

Cancer-related markers	
Membrane	E-cadherin PanCK PI3KCA MSH2 MSH6 MLH1 PMS2 ARID1A PARP1 Ki67 IDO1 GAL3
DNA mismatch repair	
Cycling	

Other markers	
PR	CD47
Immune interaction	PD1 PDL1 IFNG
Antigen presentation	CD40 HLA-DR
Nuclear	DAPI

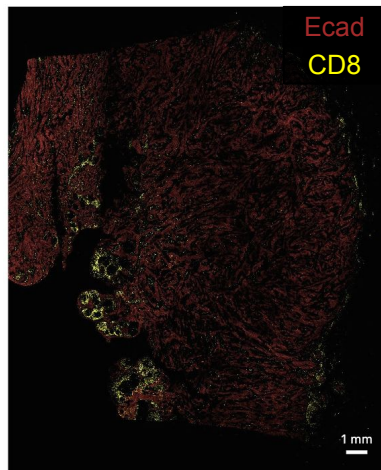


DAPI	Nucleus
E-cadherin	Tumor
HER1	Drug targets
HER2	
HER3	
HER4	
PanCK	Stroma
aSMA	
CD59	
CCNE1	
MUC4	
HIF1a	Hypoxia
Ki67	Cycling
LAMP2	Lysosomal function
SLC46A3	
SERPINE1	
CD31	Endothelial
CD45	Immune
CD3	T cell
CD4	CD4+ T cell
CD8	CD8+ T cell
CD56	NK cell
FOXP3	
Granzyme B	T cell activation
CD28	
CTLA4	
LAG3	
PD1	
CD14	Monocyte
CD16	Non-classical
CD68	Macrophage
CD163	M2-like
SPP1	
CD34	Endothelial

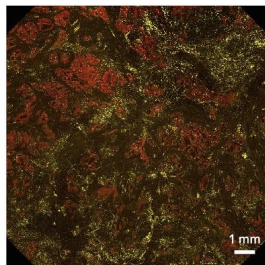
Study-to-study data variation in spatial proteomics analysis

Capture areas

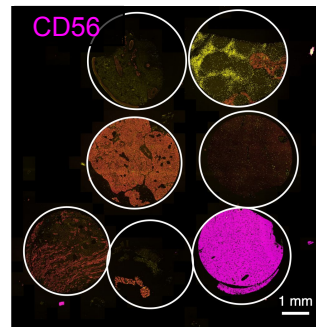
Whole slide



Targeted
capture



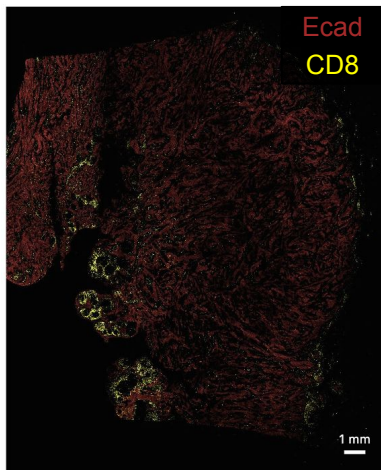
Tissue
Microarray



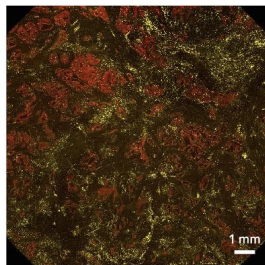
Study-to-study data variation in spatial proteomics analysis

Capture areas

Whole slide

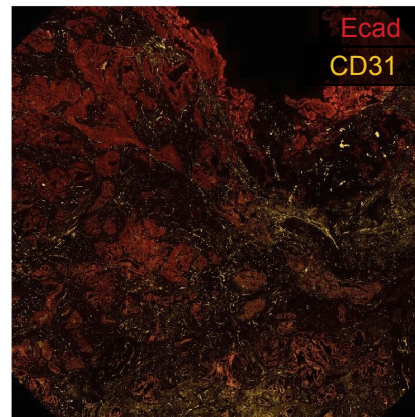


Targeted capture

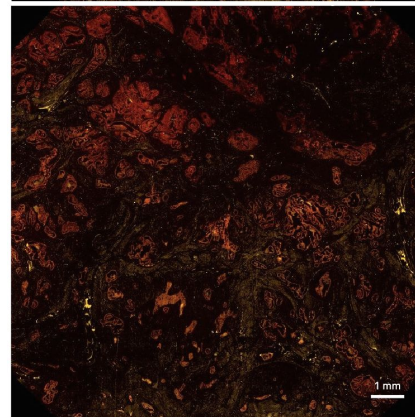


Marker failures

Marker succeeds



Marker fails



Overview: Clinical applications of segmentation-based spatial proteomics

- Segmentation converts pixels to biologically meaningful cells
- Segmented cells lead to interpretable results through straightforward analysis methods
- Segmentation-based methods are useful in studies where data is limited
 - Like in some clinical disease studies with small sample sizes
- These results can be further correlated to disease states
- Limitations - Study-to-study variation can limit pipeline generalizability
 - Differences in data generation methods, study design, and marker panel quality



Characterization of tumour heterogeneity through segmentation-free representation learning on multiplexed imaging data

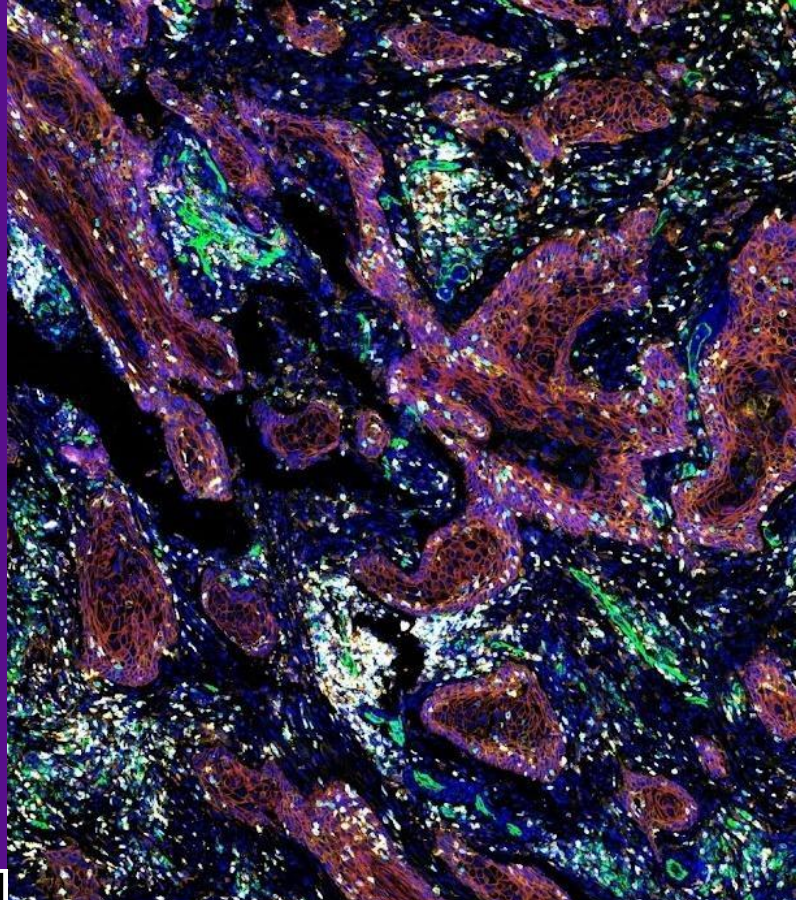
Jimin Tan

Institute for Systems Genetics
Department of Biochemistry & Molecular
Pharmacology

NYU Grossman School of Medicine



github



Section presented by David Fenyő

Our lab's focus: characterizing the tissue microenvironment

Segmentation-based approach

Cells

Relies on accurate cell-segmentation algorithm

Can work with smaller datasets, though this may limit statistical power

Cell morphologies are discarded/summarized into a shape, and marker expressions are assigned to cells

More straightforward and easier implementability (cell-centric interpretation is intuitive)

Segmentation-free approach

Unit of analysis

Image pixels / block of pixels

Preprocessing

Learns directly from raw data (often via self-supervised learning); the training has high computational costs

Data requirements

Large amounts of data is required

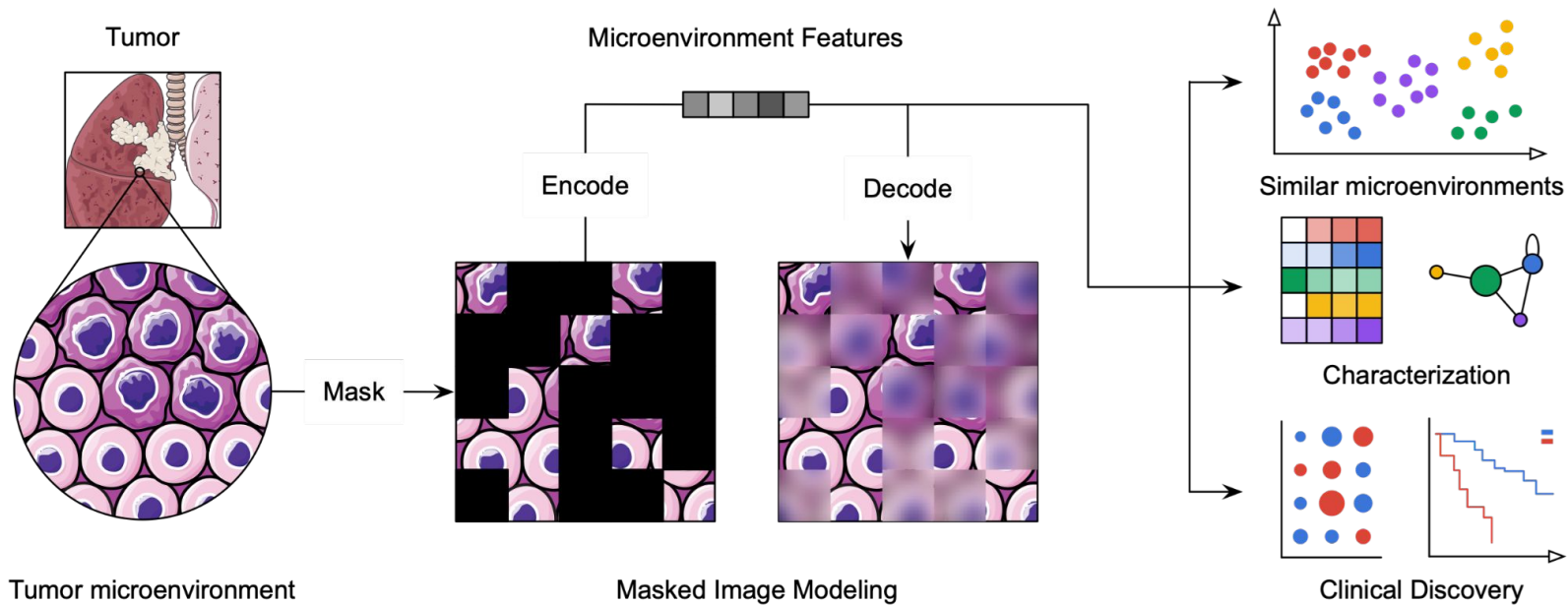
Biological information retained

Retains cell morphology and assesses marker distribution without segmentation bias

Interpretability / practical usability

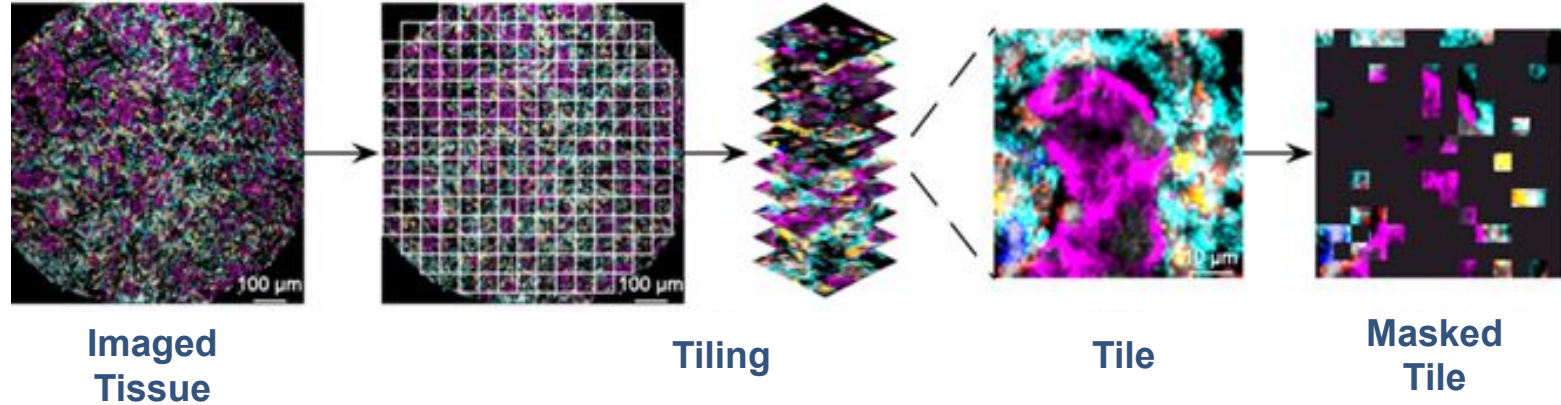
More “black box”: learned signatures are harder to interpret and may be harder to translate into clinical routine

Self-supervised learning



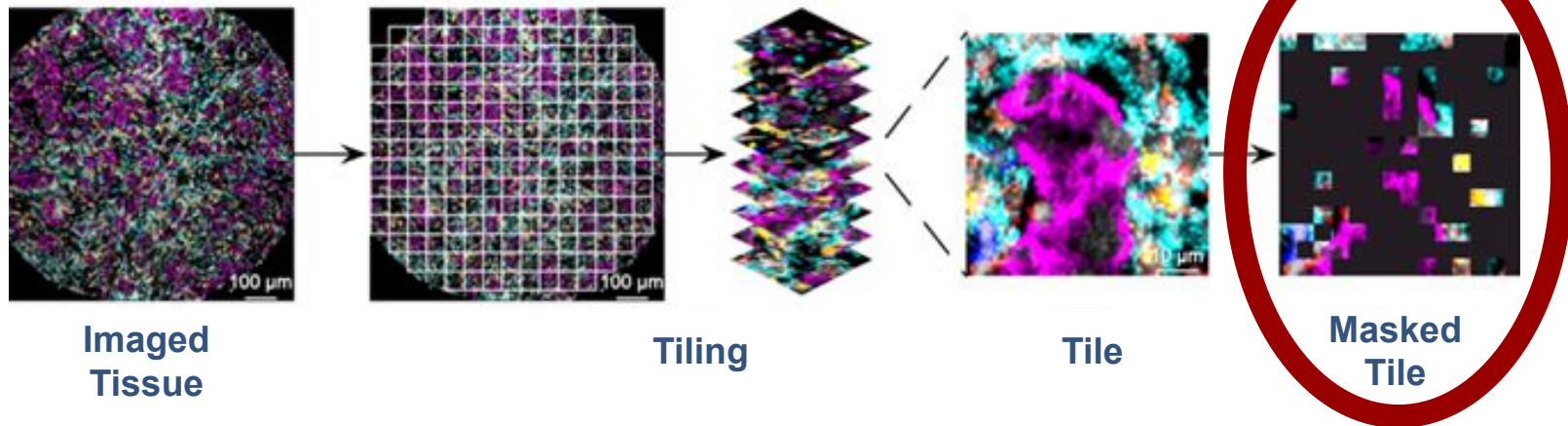
CANVAS is a vision transformer that directly takes high- dimensional multiplexed images and is trained using self-supervised masked image modeling.

Preparation



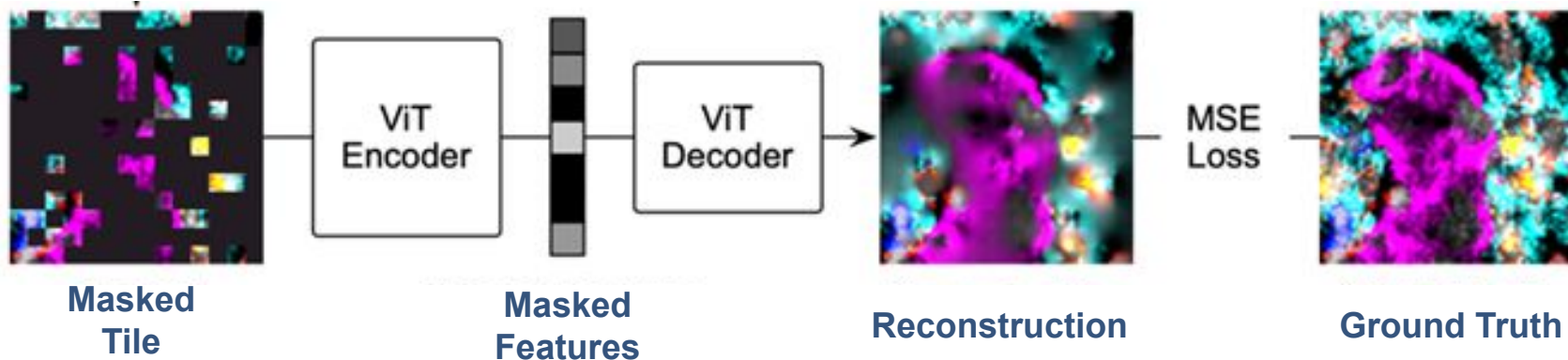
CANVAS is a vision transformer that directly takes high- dimensional multiplexed images and is trained using self-supervised masked image modeling.

Preparation

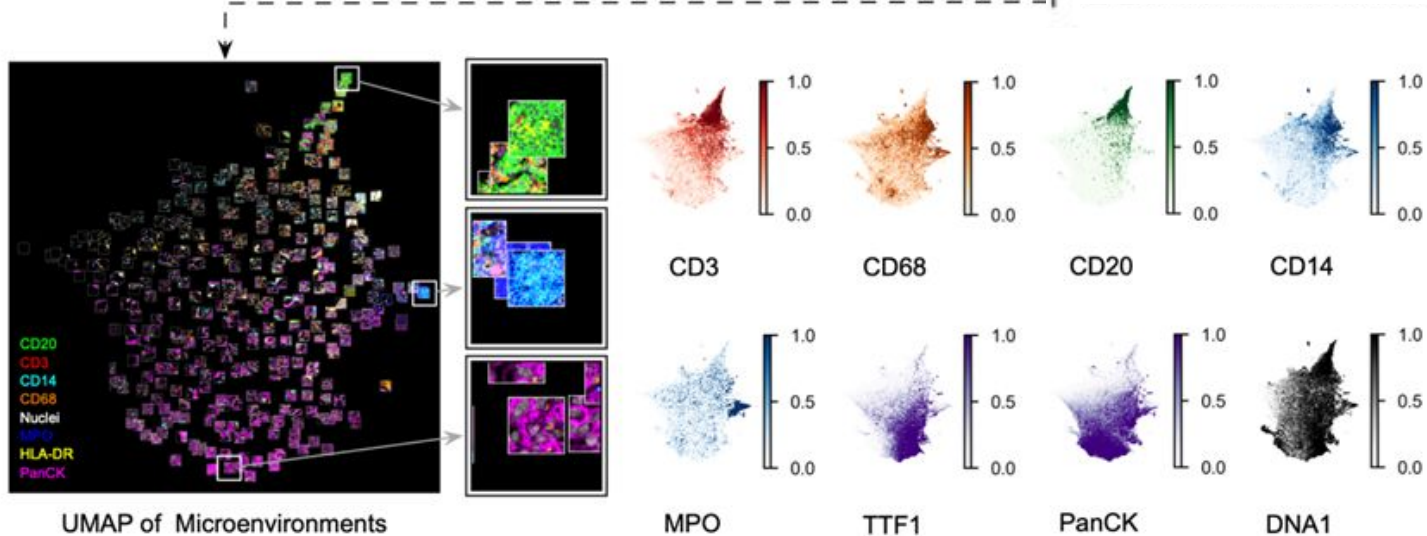
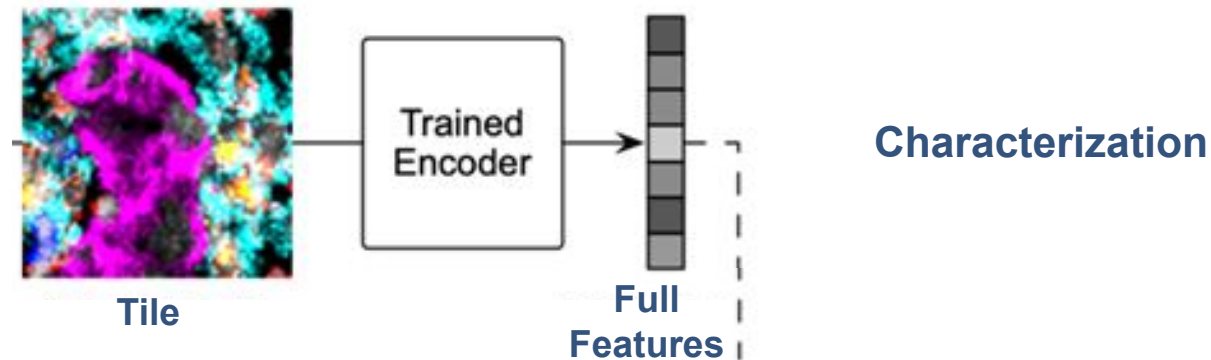


CANVAS is a vision transformer that directly takes high- dimensional multiplexed images and is trained using self-supervised masked image modeling.

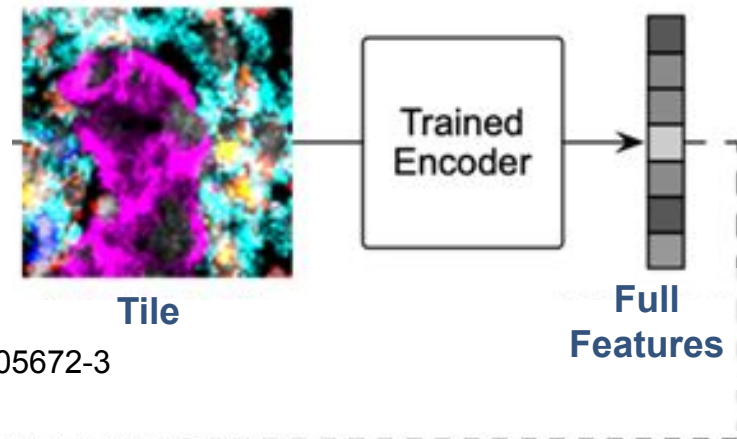
Training



CANVAS



Clustering reveal signatures



Tile

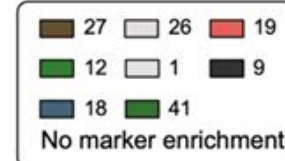
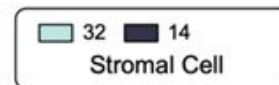
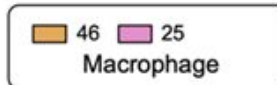
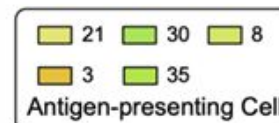
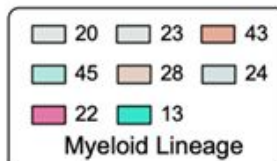
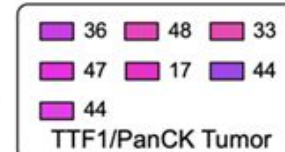
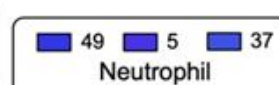
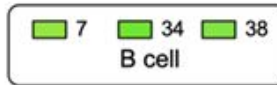
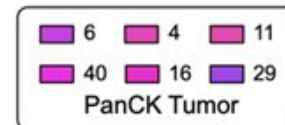
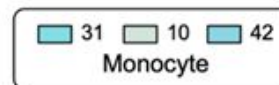
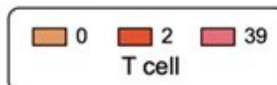
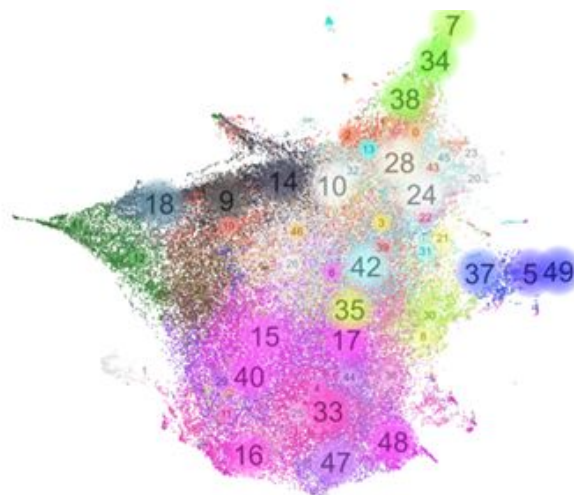
Full
Features

Data Source:

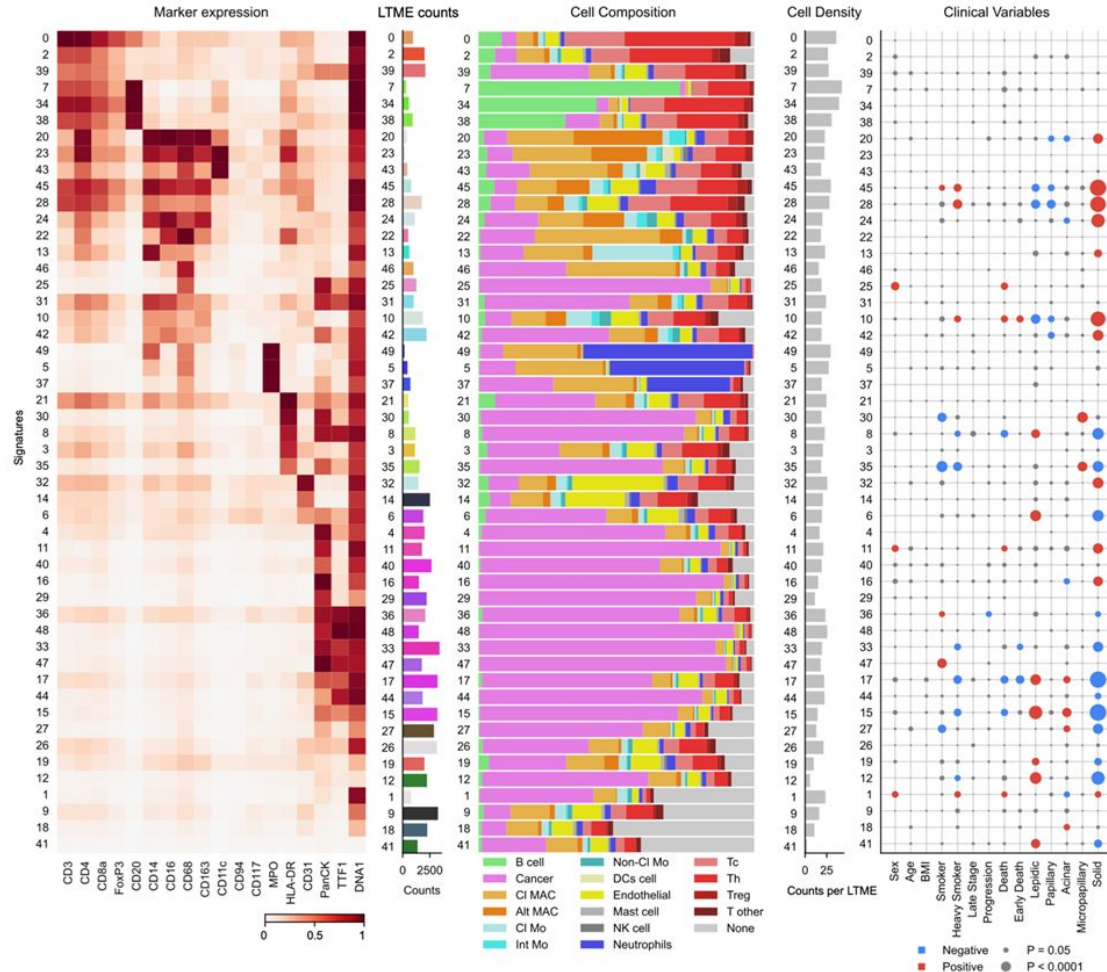
Sorin et al. Nature 2023

<https://doi.org/10.1038/s41586-022-05672-3>

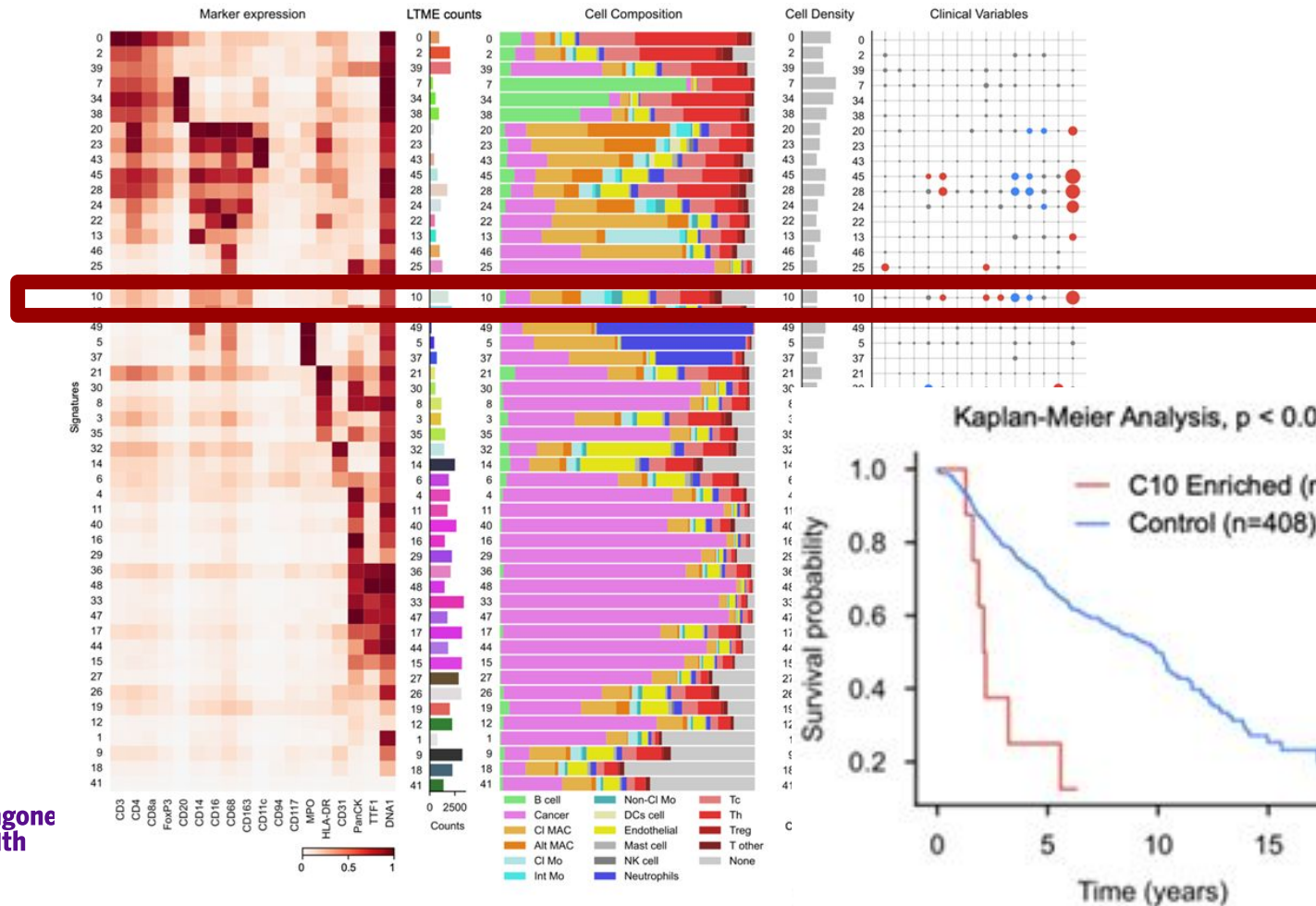
<https://zenodo.org/record/7760826>



Characterization of the signatures



Signature 10 is associated with short survival





github

SimSpace

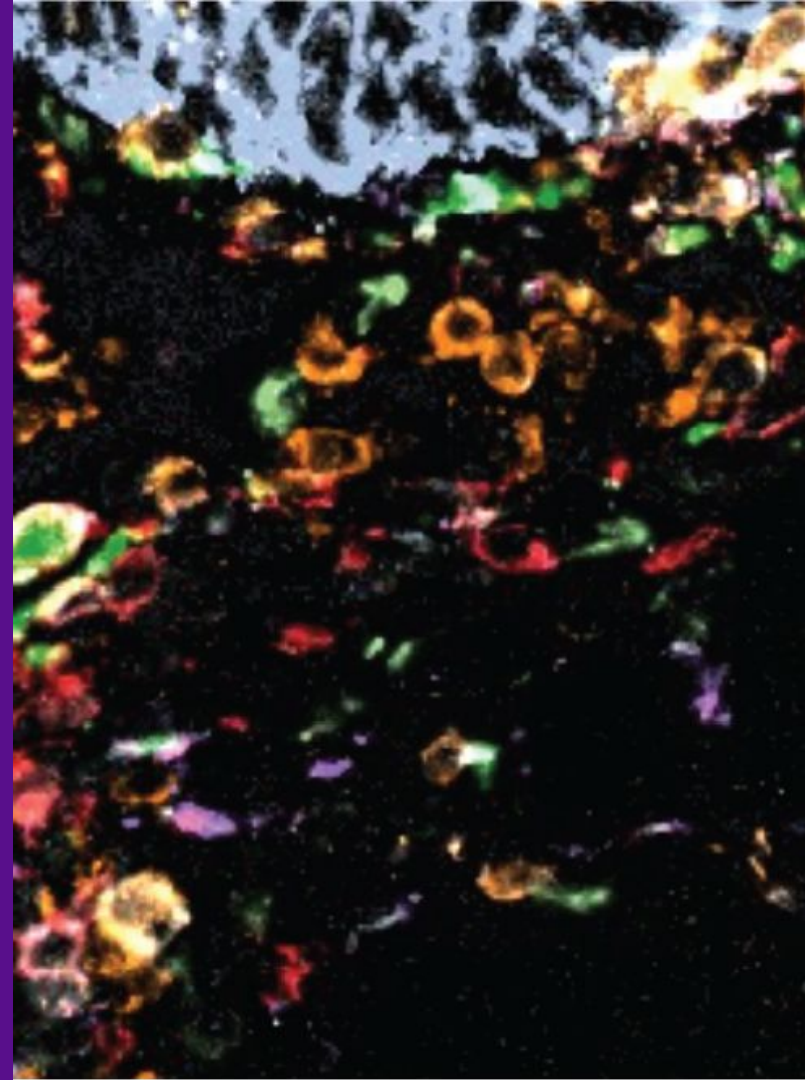
Simulated tissues for spatial proteomics

Tianxiao Zhao

Institute for Systems Genetics

Department of Biochemistry & Molecular Pharmacology

NYU Grossman School of Medicine



Ground truth for spatial analysis

**Does an analysis recover
the pattern we put in?**

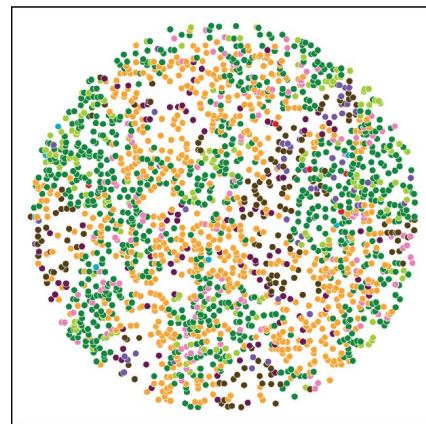
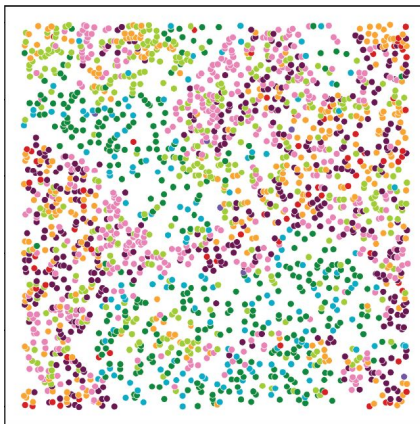
Known answer

Cell types and spatial organization

Controlled variations

Neighborhood mixing and tissue shape

Two simulated tissues



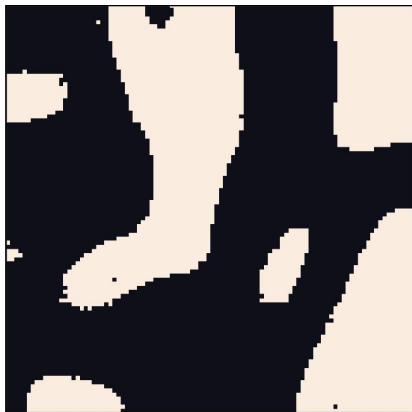
Each dot is a cell; color shows cell type

How SimSpace builds a virtual tissue

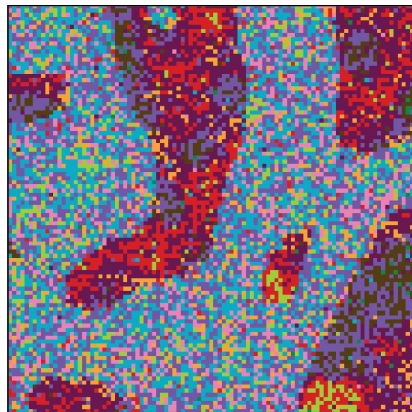
Reference-free: choose parameters

Reference-based: fit a real tissue

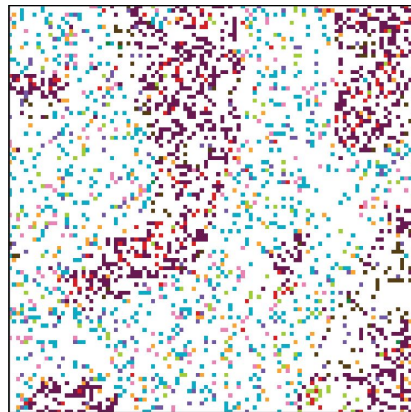
Tissue regions



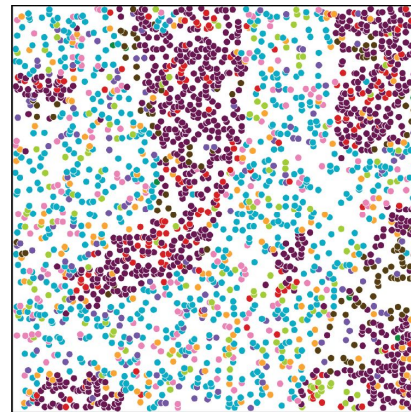
Cell types



Cell density



Coordinates



Add molecular profiles for each cell type

Output: cell positions, cell-type labels and protein measurements

A CODEX tumor becomes a simulation

Human endometrial tumor

206 reference cells

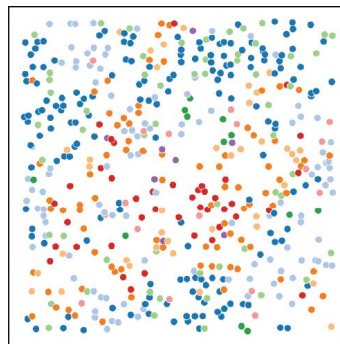
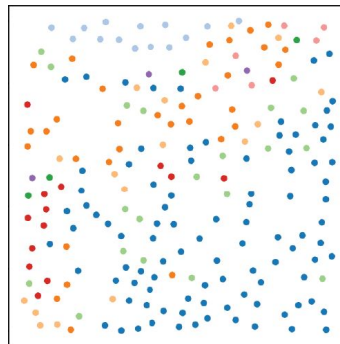
9 annotated cell types

- Tumor cells
- Stromal cells
- T cells (other)
- Endothelial cells
- Cytotoxic T cells
- CD4+ T cells
- Macrophages (CD163-)
- Macrophages (CD163+)
- Neutrophils

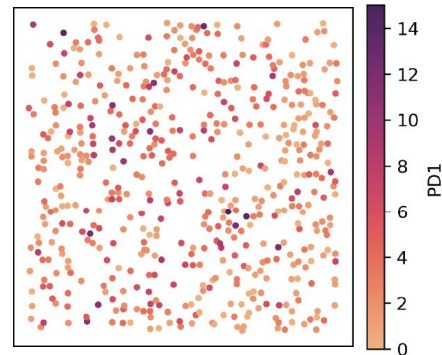
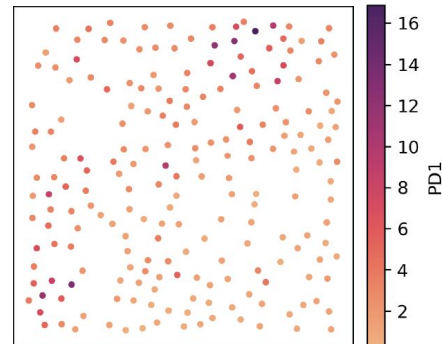
Reference

Simulated

Cell types



PD1 intensity



Virtual perturbations of T-cell affinity

Change the neighbor preference between two T-cell populations

Cytotoxic T cells

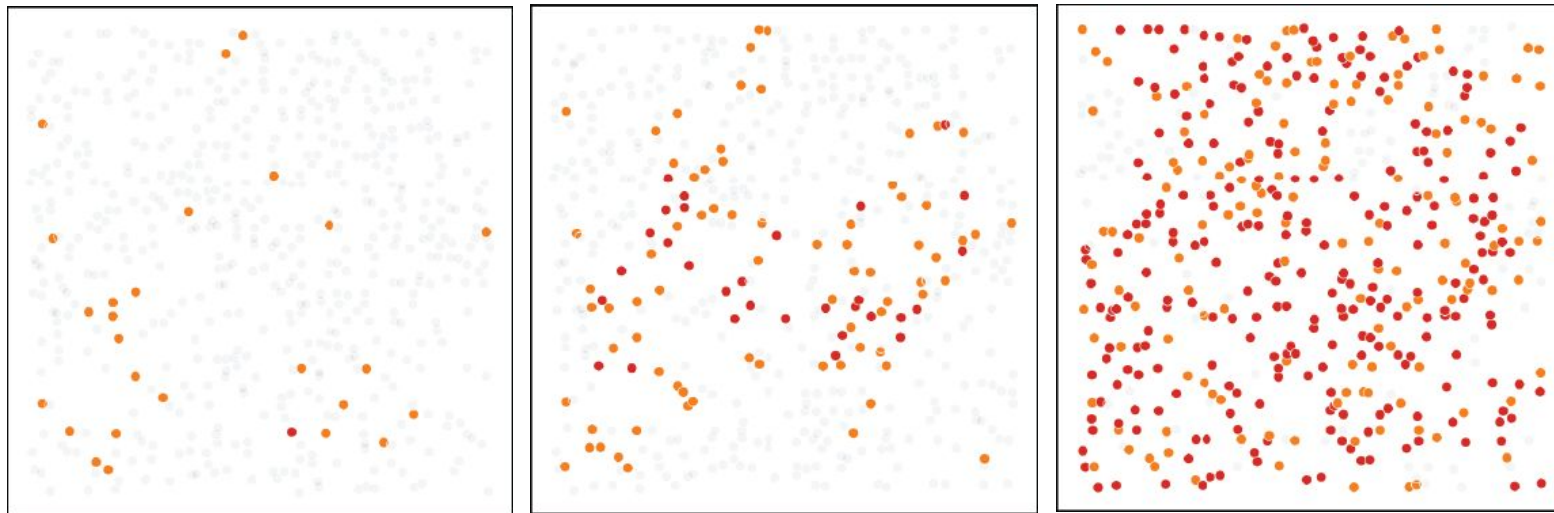
CD4+ T cells

Other cells faded

Lower affinity

Intermediate

Higher affinity



Simulated scenarios can change both proximity and cell abundance.

SimSpace manuscript, Fig. 6C, selected examples 1, 4 and 7

Your turn in Google Colab

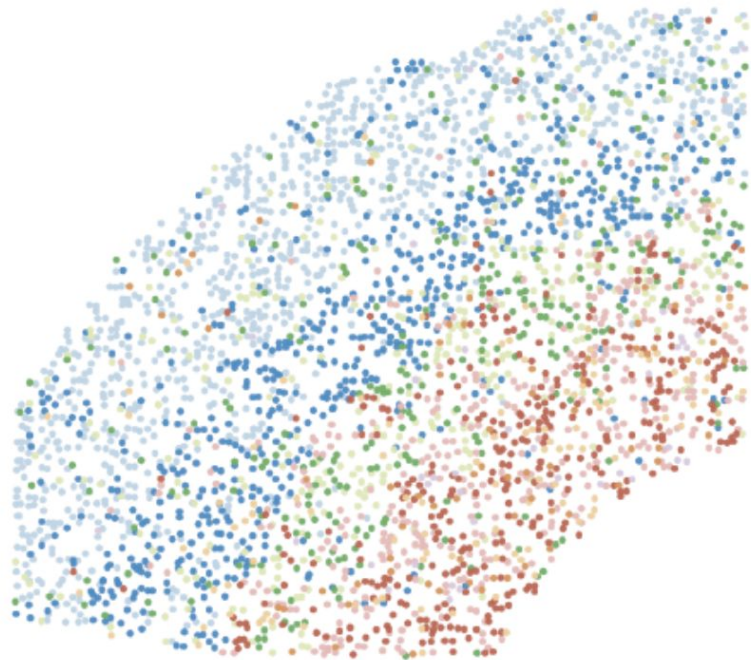
A short hands-on session

- 1 Generate a virtual tissue**
- 2 Change a T-cell affinity**
- 3 Download your simulated data**

[Open the workshop notebook](#)

github.com/TianxiaoNYU/simspace

A tissue you can change





SimSpace Demo

[github.com/FenyoLab/
hupo26-spatial-proteomics-hub](https://github.com/FenyoLab/hupo26-spatial-proteomics-hub)



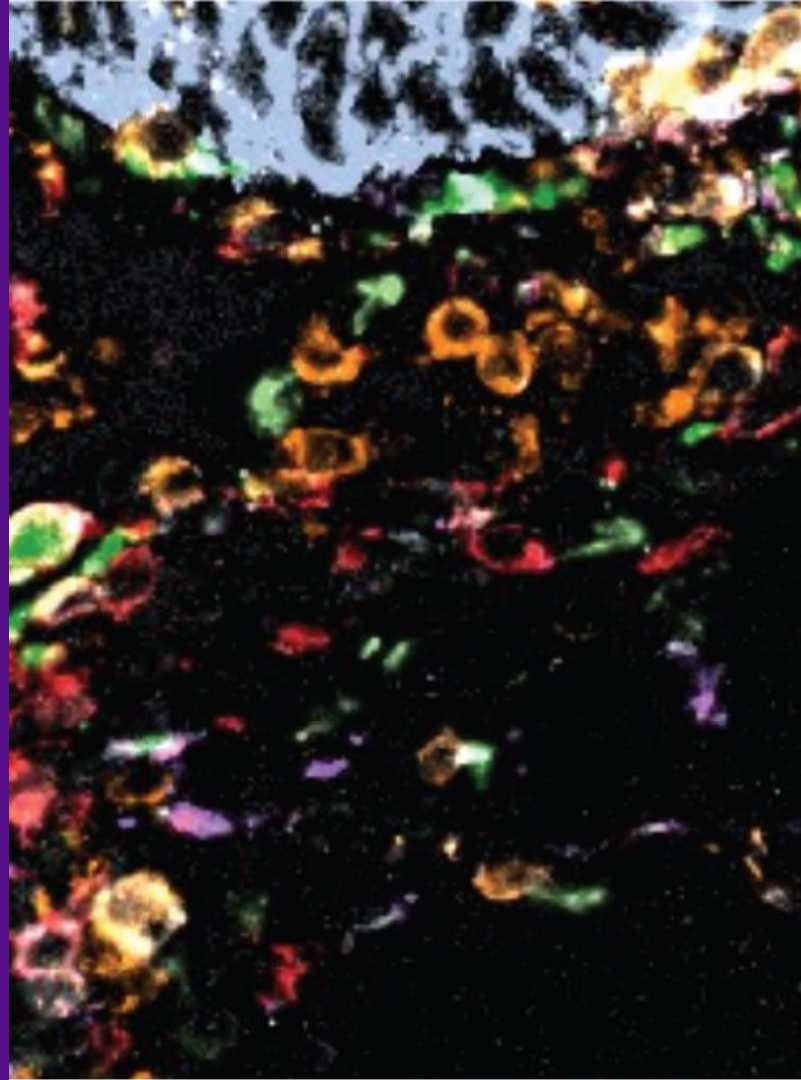
Access Simspace demo here!

Tianxiao Zhao

Institute for Systems Genetics

Department of Biochemistry & Molecular Pharmacology

NYU Grossman School of Medicine





Thank you!

David.Fenyo@nyulangone.org

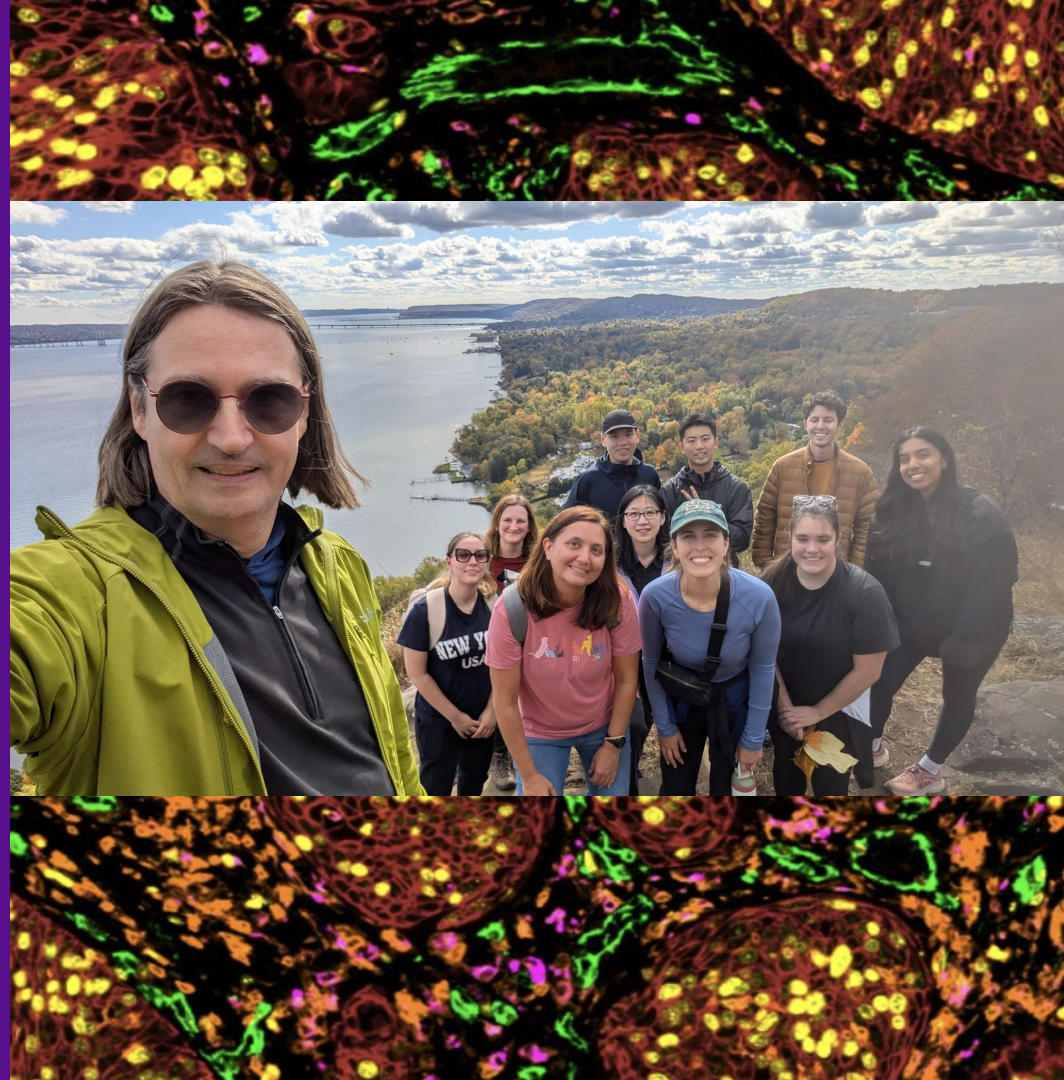
Linda.Procell@nyulangone.org

Tianxiao.Zhao@nyulangone.org

Beata.Szeitz@nyulangone.org



github



NYU Grossman School of Medicine